



Holistic quantified-self for context-aware wearable augmented reality

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ABSTRACT

We propose Holistic Quantified-Self for wearable Augmented Reality (HQS-AR), a unified user model for users wearing AR glasses that captures multi-dimensional user states through sensors and interaction data. Despite the growing need for a personalized AR system, unified user modeling in wearable AR remains limited, often leading to suboptimal experiences. To address this gap, we identified key modeling dimensions and their corresponding factors: physical (posture, energy expenditure), cognitive-emotional (cognitive load, emotion), social (social engagement, interpersonal density), and behavioral (device type, usage log). Using motion, visual, audio, behavioral, and physiological datasets, we implemented machine learning models to infer user states and classify scenarios in office environments, achieving an average accuracy of 88% for state estimation and 99% for scenario classification in hold-out validation. Our user model is expected to serve as a state indicator for configuring wearable AR systems and a tool for modeling user routines, advancing personalized AR interfaces.

1. Introduction

Augmented reality (AR) seamlessly overlays virtual content onto the user's real-world environment, creating interactive experiences (Azuma, 1997). As wearable AR devices—such as AR glasses—become more prevalent, there is an increasing demand for personalized, context-aware AR experiences that adapt to users' current needs (Lindlbauer, 2022), extending beyond explicit indicators (e.g., environment, spatial information). For example, adjusting AR content based on the user's load or usage patterns can reduce distractions and enhance user experience. Delivering such personalized experiences requires a comprehensive understanding of the user's states and behaviors, which can be represented as a user model (Kobsa, 1993; Cena et al., 2018).

Achieving effective personalization in AR requires a domain-specific user model, yet existing approaches have limitations. First, although integrated user models have been proposed in web and ubiquitous computing environments (e.g., Cena et al., 2019; Musto et al., 2020), many are not optimized for the unique demands of AR. Second, while several adaptive AR systems consider specific user factors (e.g., Davari et al., 2022), multidimensional and dynamic profiling for optimal AR delivery remains underexplored. Third, while multimodal datasets have been explored for user state classification (e.g., Bhat et al., 2023), many rely on additional, less practical sensors (e.g., scalp- or chest-worn), or lack consideration for real-world constraints, making them less suitable for wearable AR.

As a result, there is a growing need for an AR-specific user model that integrates the most relevant user dimensions, combines AR glasses' multimodal data with existing inputs, and demonstrates its applicability in real-world AR scenarios within a unified framework. Because AR glasses can capture not only the user's first-person visual and auditory streams but also IMU and head movement data, they offer deeper contextual insights than conventional desktop PC- or smartphone-based multimodal sensing. By bridging this gap, researchers and designers can illustrate how a holistic user model operates in real-world AR, ultimately enhancing responsive interface adaptation.

Therefore, we present a unified user model for wearable AR, identifying critical dimensions and key factors for understanding the user and configuring the AR interface. Our main contributions are as follows:

- We propose Holistic Quantified-Self for wearable AR, a unified user model that integrates the user's physical, cognitive-emotional, social, and behavioral dimensions with key factors, grounded in theoretical and practical relevance for wearable AR.
- We develop methods to infer posture, energy expenditure, cognitive load, emotion, social engagement, and office scenarios, leveraging a multi-sensor dataset encompassing motion, visual, audio, behavioral, and physiological data.

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- We demonstrate the notable accuracy of our estimation models across four distinct validation methods, achieving reliable estimation within an office environment.
- We present practical application scenarios, illustrating how state indicators are used to configure virtual interfaces and how recurring state patterns model user routines to enhance contextual relevance and personalization of AR experiences.

By continuously monitoring users to dynamically adapt what, where, how, and when information is presented, our user model lays the foundation for personalized AR experiences. For example, by detecting transitions and fluctuations in user states, the system can control notifications (e.g., delay non-urgent alerts during high load), recommend content (e.g., based on user fatigue), and adjust interaction modalities (e.g., from gesture to voice) to prevent overload, reduce distractions, and avoid conflicts—ultimately addressing users' implicit needs. Beyond extending multimodal sensing, this work presents real-world applicability of AR-specific user model, supporting user-centered augmentation.

2. Related works

2.1. Context-aware mixed reality

Context-awareness, the ability of a system to sense and react to its environment and the user, is a fundamental concept in ubiquitous computing (Dey, 2001). The AR environment, with its ability to continuously sense both the physical world and the user, offers significantly more opportunities for context-aware adaptation than traditional computing settings. Pervasive AR (Grubert et al., 2017) further develops this concept, ensuring continuous and unobtrusive adjustments to the AR interface. As context-aware AR systems offer personalized experiences by responding to changing user needs and situations (Grubert et al., 2017; Minh Tran et al., 2023), a number of tools have been developed to support such adaptation, including CAPtURAR, an AR authoring tool for human-involved context-aware applications (Wang et al., 2020), and adaptive user interface toolkits (AUIT) (Evangelista Belo et al., 2022).

Recognizing context is a critical prerequisite for enabling context-aware adaptation. Recent studies have explored multimodal sensing in various devices to infer environmental and user-related context. However, user state inference still remains in its early stages compared to environmental sensing in XR environments (Xu et al., 2023). Outside AR, prior work has explored various sensing devices for user context recognition, including smartphones (Otebolaku and Andrade, 2016), smartwatches (Laput et al., 2016), earables (Röddiger et al., 2022), and VR systems (Souchet et al., 2022). Compared to these, AR smart glasses offer unique advantages—such as a continuous, first-person perspective and the potential for multi-sensor integration—enabling a richer and more contextually grounded understanding of the user.

Building on this potential, several studies have used custom-built glasses with diverse sensors for facial expression recognition (Yan et al., 2022), emotion inference (Zhao et al., 2022), stress detection in law enforcement scenarios (Karakostas et al., 2024), and the collection of physiological signals to support attention and cognitive state estimation (Chwalek et al., 2021). More recently, Kim et al. (2025) proposed Explainable XR, an LLM-assisted framework for interpreting XR user actions, though designed for offline analysis rather than interface adaptation. Taken together, prior work (1) relies on non-commercial or custom-built glass hardware unsuited for daily use; (2) is confined to specific scenarios; (3) lacks integration strategies with adaptive AR interfaces; or (4) focuses on signal acquisition over state modeling. In contrast, we address the full pipeline: from general-purpose data collection and user state inference to its integration scheme into adaptive AR interfaces using commercially available smart AR glasses in everyday scenarios.

2.2. User-centered adaptations in mixed reality

Mixed reality (MR) adaptation can be categorized as environment-centered (Cheng et al., 2021; Qian et al., 2022) or user-centered. This paper focuses on user-centered adaptation, which tailors AR experiences to the user's state, even within the same physical space. Prior studies have explored user-centered adaptation, often targeting a specific factor. For example, researchers have explored optimal virtual content placement during social interaction (Orlosky et al., 2015; Casoria et al., 2022; Davari et al., 2022), effective interface design while moving or walking (Tsai and Huang, 2018; Lages and Bowman, 2019a,b), or adaptive amount of information based on cognitive load (Lindlbauer et al., 2019). Despite demonstrating the value of user-centered adaptation, real-world AR scenarios involve multiple and dynamic combinations of user states, requiring a more comprehensive adaptation strategy.

To address this, some prior research has attempted to combine multiple contextual cues, primarily in handheld AR settings. Handheld AR involves mobile devices held in the user's hand, such as smartphones or tablets, whereas wearable AR refers to head-mounted displays or smart glasses worn on the body. For instance, Hervás et al. (2013) employed ontologies (user, device, physical environment, service) to personalize handheld AR, and Chatzopoulos and Hui (2016) considered user's focus, mood, social network, time, and location for context-aware recommendations on handheld AR. While these approaches incorporate diverse contextual cues, they may not address key factors that shape interface design in wearable AR. Because the determinants of interface design differ between handheld screens and head-mounted displays, there is a need to identify theoretically grounded contextual dimensions specific to this medium and to establish corresponding principles for adaptive interface design.

Although relatively limited, there have been attempts to integrate multiple factors for wearable AR. Stefanidi et al. (2022) optimized interface configurations such as level of detail and UI component prioritization based on user profile, state, environment, and task. More recently, Janaka et al. (2024) proposed TOM, a development platform that enables context-aware service in wearable intelligent assistants by understanding users' physical, cognitive, and affective states in use cases like running. However, these approaches (1) lack generalizability due to reliance on specific scenarios and indirect inference of user states, and (2) lack a theoretically grounded framework for systematically identifying and modeling user states that drive adaptation. This work addresses these limitations by proposing a structured framework and inference methods for wearable AR adaptation beyond ad-hoc use of available sensor data in everyday scenarios.

2.3. User modeling in MR

For a personalized system to effectively adapt itself to satisfy individual users' needs, a user model, a digital representation of a user, is essential (Kobsa et al., 2001; Brusilovsky, 2007; Cena et al., 2019). In web and ubiquitous computing, integrative user models have been widely explored. For instance, Jang et al. (2005) developed a 5W1H-based unified user-centric context model with static contextual data to enhance situation recognition and automation in smart environments. While Cena et al. (2019) proposed a hierarchical user model that infers short-term user states from sensor data and derives long-term behavioral patterns, Musto et al. (2020) introduced a holistic user model integrating digital footprints such as demographics, interests, psychological traits, and behaviors to support personalized services.

In contrast, in the AR domain, the development and integration of user models have been relatively underexplored. For example, efforts such as context-aware mobile augmented reality (CAMAR) (Oh et al., 2009) enable personalized information delivery and selective content sharing, while user movement model (Makolkina and Paramonov, 2020) optimizes data request mechanisms based on movement,

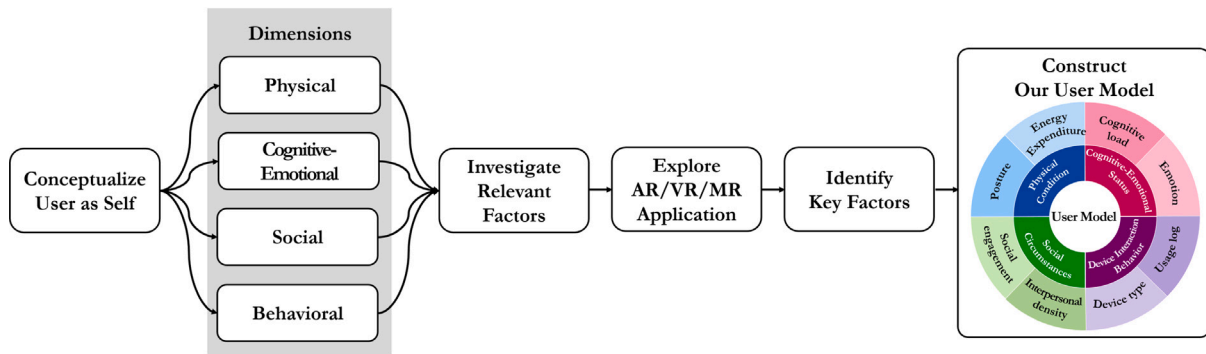


Fig. 1. Designing process of our user model.

enhancing network efficiency and real-time AR content delivery. Other studies focused on specific applications, such as AR tour trajectory-based user model (Shakeri et al., 2023) for personalized AR content, AR-game-experience model (Shin, 2019) integrating cognitive and motivational constructs, and user profiling based on personality traits and demographics (Lu et al., 2023; Tricomi et al., 2023). More recently, Li et al. (2025) proposed a belief-desire-intention-based user model for adaptive systems focused on procedural task guidance.

While these studies contribute to AR personalization, they often focus on (1) handheld AR in some cases, (2) specific user attributes (e.g., movement) or contexts (e.g., tourism, gaming), or (3) static profiling or procedural tasks. However, wearable AR demands a more comprehensive and dynamic user representation—one that captures factors directly affecting interface configuration and interaction. To address these limitations, we propose a user model tailored to AR glass users, integrating key factors across broader scenarios with an emphasis on dynamic profiling. This model aims to serve as a reference framework for enhancing personalized AR experiences through responsive and adaptive interfaces.

3. Method

This study develops a holistic user state model for wearable AR interfaces, addressing the following research questions:

- **RQ1:** Which user factors should be considered when developing a user model for wearable AR interfaces?
- **RQ2:** How can the identified user factors be estimated using multimodal sensor data in an AR scenario?
- **RQ3:** What considerations are essential for developing future user-adaptive AR systems?

To address **RQ1**, we explored prior work to identify key user factors, defining the problem space and summarizing relevant literature (Section 3.1 to 3.2). For **RQ2**, we collected a multimodal dataset, developing algorithms to infer user factors (Sections 3.3 to 3.7). For **RQ3**, we gathered user feedback about the user model (Section 3.4).

3.1. Design rationale

This section outlines the design rationale for our user model. Section 3.1.1 defines the key modeling dimensions and their theoretical foundations. Section 3.1.2 presents the problem space that guided our literature survey. Section 3.1.3 summarizes the survey findings. The designing process is illustrated in Fig. 1.

3.1.1. Modeling dimensions

From the biopsychosocial model (Engel, 1979, 1981), the physical, psychological, and social aspects of the self have been central to unified models in psychology, influencing research on human functioning and augmented human, which often segments the concept of self into three dimensions: physical, mental, and social (Nilsson and Kristenson, 2010; Gobbens and van Assen, 2014; Guerrero et al., 2022). HCI research expands on these foundations by incorporating behavioral factors to better capture user contexts. For example, Harari et al. (2016) noted that smartphone data reflect daily activities, social interaction, and mobility patterns, while Cena et al. (2019) proposed user modeling including user behavior, affects, cognitive functions, physical states, and social relations.

Similarly, ubiquitous computing research has introduced digital behavior aspect as crucial dimension. Studies by Yoon et al. (2014), Lee et al. (2018a) demonstrate how physical, cognitive-emotional, social, and behavioral factors together create a richer dynamic user profile for personalized services in augmented human research. The behavioral dimension captures interaction patterns with digital contents—enabling passive monitoring of user preferences and context recognition. Building on these foundations, this work establishes these four dimensions as essential for understanding the user, extending user-model frameworks to wearable AR. A visual summary of self/user dimensions from prior works is presented in Table 1.

3.1.2. Problem space

Before the literature survey, we defined the problem space based on design space analysis and methodology by MacLean et al. (2020) and Xu et al. (2023). We categorized it into two aspects: user-focused (*who*) and configuration-focused (*what, where, how, when*).

For the user-focused aspect, we examined how the four dimensions—physical, cognitive-emotional, social, and behavioral—have been studied and which user factors have been addressed in prior work. This helped identify the core user factors relevant to each dimension.

For the configuration-focused aspect, we investigated how these user factors have been applied to MR content configuration, structured as *what* (content, level of detail), *where* (position, coordinate system), *how* (occlusion avoidance, interaction mode), and *when* (timing) for content display, drawing from prior studies (Lindlbauer et al., 2019; Xu et al., 2023). This approach helps our model be both theoretically grounded and practical for AR.

3.1.3. Key factors

This section summarizes literature addressing the problem space, classified into one or more of the following four categories: (1) user-focused studies (*who*), examining dimension and its factor conceptualization; (2) configuration-focused studies (*what, where, how, when*), examining its influence on virtual interface design; (3) studies highlighting the importance of incorporating specific user factors into general system design; and (4) studies on sensing and measuring user factors in MR.

Table 1

Comparison of self/user modeling dimensions across prior work and our approach.

Prior work	Physical	Psychological	Social	Behavioral	Domain area
Engel (1979, 1981)	✓	✓	✓		Behavioral Science
Gobbens and van Assen (2014)	✓	✓	✓		Quality of Life
Guerrero et al. (2022)	✓	✓	✓		Augmented Human
Harari et al. (2016)	✓		✓	✓	Psychology
Cena et al. (2019)	✓	✓	✓	✓	UbiComp
Yoon et al. (2014)	✓	✓	✓	✓	Augmented Human
Ours	✓	✓	✓	✓	Wearable AR

Table 2

Overview of the conceptualization of dimensions from prior theories, models, and frameworks.

Dimension	Component	Sub-component (Level)	Related theories, models, frameworks
Physical Condition	Energy Expenditure	Physical activity (>1.5 METs ^a)	Terminology consensus project (Tremblay et al. (2017))
		- Vigorous PA	
		- Moderate PA	
		- Light PA	
	Posture	Sedentary behavior (≤ 1.5 METs)	
		Sleep (<1 MET)	
		Standing	
		Sitting	
		Reclining	
Cognitive-Emotional Status	Cognitive Load	Cognitive load (Low/Medium/High)	Cognitive load theory (Sweller (1994))
	Emotion	Arousal (Low/Medium/High)	Circumplex model of affect (Russell (1980))
		Valence (Negative/Neutral/Positive)	
Social Circumstance	Social Interaction	Actor	6 components of social interaction (APRACE) (Hoppler et al. (2022))
		Partner	
		Relation	
		Activities	
		Context	
Device Interaction Behavior	Device Type	- Desktop PC - Laptop/Tablet - Smartphone	Device ontology (Hervás et al., 2013)
	Usage Log	- App use log	Cyber activity pattern discovery framework (Li et al., 2022)

^a MET = metabolic equivalent.

A total of 48 papers were selected from MR, HCI, and Psychology, spanning publications from 1980 to 2023. Table 2 provides an overview of how each dimension is structured, while Appendix Table A.1 summarizes how prior studies relate to the user components and parameters defined in this study. The discussion first provides an overview of each dimension, followed by an analysis of specific factors and their role in system design.

Physical condition. In UI layout, physical condition refers to a user's external bodily state, structured by *energy expenditure* and *posture*. Tremblay et al. (2017) proposed a conceptual model, defining movement-based states with the combination of energy expenditure (e.g., sleep, sedentary behavior, physical activity) and posture (e.g., lying, sitting, standing). This framework provides a structured classification of physical states. Monitoring physical states is crucial as they influence content relevance, detail, placement, occlusion avoidance, and timing in AR.

Content Relevance and Detail (what) A user's bodily state influences content relevance and complexity in AR. For example, iDisplay (Tsai and Huang, 2018) adjusts content and its level of detail to mitigate cognitive overload during movement, while an AR content recommendation model by Shakeri et al. (2023) personalizes content

based on spatial behaviors. These findings suggest that adapting content and detail based on physical state can reduce cognitive load and enhance personalization.

Content Positioning and Coordinate Systems (where) AR content placement should adjust to physical states for real-world usability. To prevent visual obstruction, studies explored moving focus modes (Tsai and Huang, 2018), adjustments in coordinate systems (Lages and Bowman, 2019a,b), and placement for different body positions (seated vs. standing) (Cheng et al., 2023).

Research on text positioning (Klose et al., 2019) and coordinate systems (Fukushima et al., 2020) showed their impact on readability while walking. To improve accessibility, Glanceable AR (Lu and Bowman, 2021) and spatial placement strategies (Lee and Woo, 2023; Lee et al., 2023a) were developed. These findings suggest adapting placement based on physical states can improve AR usability.

Occlusion Avoidance (how) and Timing Strategies (when) To prevent visual interference and optimize AR experience, AR should adapt to user movement. iDisplay ensures clear visibility by reducing

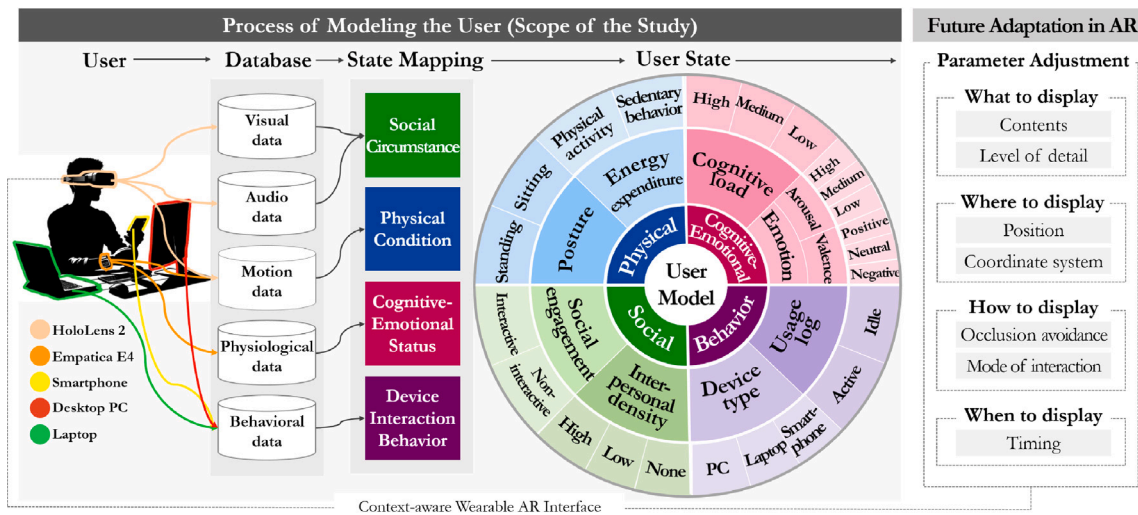


Fig. 2. Overview of the *Holistic Quantified-Self for wearable AR* framework. Multimodal data are mapped across four dimensions: social circumstances, physical condition, cognitive-emotional status, and device interaction behavior. These dimensions form the user model, guiding AR adaptations for a context-aware wearable AR interface.

obstructions (Tsai and Huang, 2018), while AR user movement modeling aims to enhance experience by delivering timely content (Makolkina and Paramonov, 2020). These findings suggest that adjusting visibility and timing helps minimize real-world disruption and enhance user immersion.

Cognitive-emotional status. Cognitive-emotional status refers to the user's internal mental state. While *cognitive load* (CL) represents the mental effort required for a task (Sweller, 1988, 1994; Hollender et al., 2010), *emotions* are mapped within a two-dimensional space of *arousal* and *valence*, based on the circumplex model of affect (Russell, 1980; Russell et al., 1989). These factors are critical, as CL affects content complexity, and emotions influence content personalization in AR.

Cognitive Load and Level of Detail (what): Monitoring cognitive load helps adjust content detail to prevent overload. Studies have explored information filtering, level of detail, and adaptive density techniques (Julier et al., 2000; DiVerdi et al., 2004; Tatzgern et al., 2016) to reduce visual clutter, while Lindlbauer et al. (2019) adjusted detail levels based on CL. These findings suggest that tailoring the detail level based on the user's load can reduce the risk of overload.

Emotion and Content (what): Emotion-aware AR systems enhance content personalization, as seen in mood- or emotion-integrated AR prototypes and recommendation systems (Chatzopoulos and Hui, 2016; Ronquillo et al., 2020), and broader adaptive and recommendation systems (Winoto and Tang, 2010; Ayata et al., 2018; Alipour et al., 2023). In explainable AI for AR, emotions are considered part of user state (Xu et al., 2023), supporting user-aware interactions. Findings suggest emotion-aware AR can enhance personalization and empathetic AR interaction, responding to user needs (Minh Tran et al., 2023).

Social circumstances. We define social circumstances as two aspects that are crucial for context-aware AR: *social engagement* (whether the user interacts with others) and *interpersonal density* (the number of nearby individuals), both of which can be inferred using implicit methods (Gauch et al., 2007; Eke et al., 2019). As APRACE (Hoppler et al., 2022) further decomposes social interaction—corresponding to social engagement—into six detailed facets, its taxonomy may inform more fine-grained AR adaptation in future work.

Understanding social contexts is essential for supporting real-world interactions while minimizing disruptions. Low engagement with low density may suggest solitary context, allowing user-focused interactions. High engagement with low density might indicate a conversation, where AR can assist communication. Low engagement with high density might reflect public settings, requiring minimal AR intrusion.

Adjusting content, placement, occlusion avoidance, and interaction mode to these contexts enhances AR usability.

Content (what) and Placement (where): Context-relevant AR content improves real-world communication, as seen in context-sensitive AR systems (Ajanki et al., 2010; Davari et al., 2022). Placement strategies should avoid obstructive overlays (Medeiros et al., 2022) and minimize interference with interactions (Orlosky et al., 2015; Janaka et al., 2022). These findings suggest that dynamic content placement upon the user's social interaction helps prevent disruption and supports social interaction.

Occlusion Avoidance and Interaction Mode (how): Reducing visual disruptions—closely related to placement decisions—is critical, as seen in occlusion-aware interfaces preventing face obstruction (Davari et al., 2022). Subtle interaction techniques, such as Itchy Nose (Lee et al., 2017, 2018b) and SelfSync (Lee et al., 2019), could enable socially acceptable interactions. These findings underscore the importance of monitoring social contexts to optimize occlusion avoidance and interaction techniques.

Device interaction behavior. Device interaction behavior refers to the users' digital activities, passively tracked through smart devices. Identifying device types is essential for seamless AR integration, enabling adaptive interactions based on available devices, as shown in device ontology (Hervás et al., 2013). The cyber activity pattern discovery framework analyzes digital behavior from app usage logs (Li et al., 2022), a primary data source for context modeling (Harari et al., 2016; Bahrainian and Crestani, 2017). In AR, understanding *device usage* and *usage log* helps optimize virtual element configuration.

Contents (what): User interaction logs, such as app usage, provide insights for personalized content adaptation, similar to clickstream analysis in web environments (Ho et al., 2011). These logs reveal preferences (Liang et al., 2019), contexts and behaviors (Lee et al., 2023b), enabling content personalization in MR (Shakeri et al., 2023). These findings suggest that adapting content based on usage logs helps identify user preferences and contextual needs.

Position (where): Adapting AR content placement across devices overcomes HMD limitations. Mobile-AR integration (Feiner et al., 1997), dual-device systems like dualCAD (Millette and McGuffin, 2016) and MARVIS (Langner et al., 2021), and smartphone-assisted AR (Bang and Woo, 2023) enhance user experiences, modeling, visualization, or reading experiences. These findings highlight the importance of tracking device usage to enable adaptive content placement across multiple displays to improve AR usability.

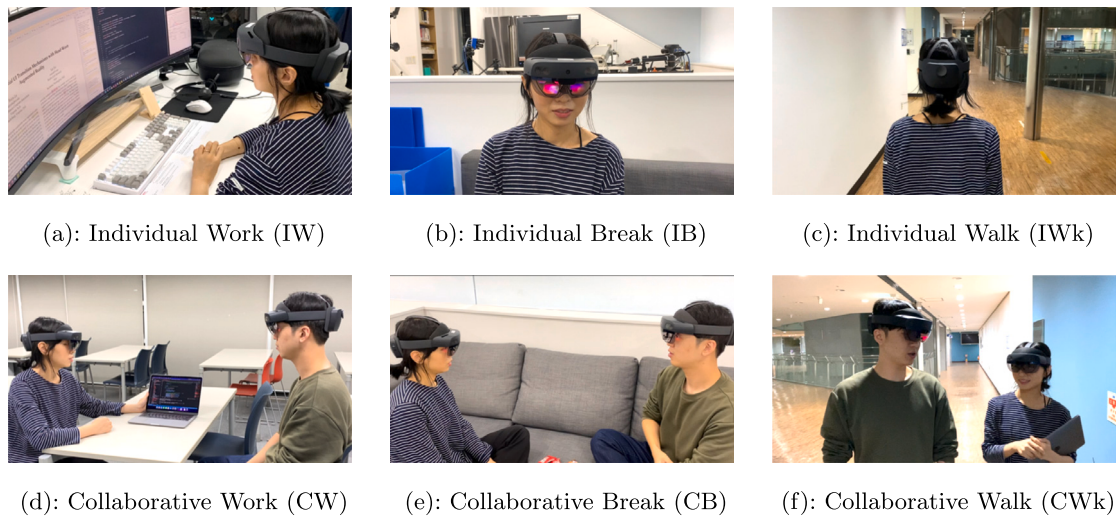


Fig. 3. The scenes from the dataset acquisition procedure.

3.2. Holistic quantified-self for wearable AR

Based on the theoretical foundations established in Section 3.1, we propose **Holistic Quantified-Self for wearable AR (HQS-AR)**. Fig. 2 illustrates the user modeling process.

- **Physical Condition:** Defined by posture (e.g., standing, sitting) and energy expenditure (e.g., physical activity, sedentary behavior).
- **Cognitive-Emotional Status:** Captured through cognitive load (e.g., high, medium, low) and emotion, represented by arousal (e.g., high to low) and valence (e.g., positive to negative).
- **Social Circumstances:** Characterized by social engagement (e.g., interactive, non-interactive) and interpersonal density (e.g., high, low, none).
- **Device Interaction Behavior:** Defined by device type (e.g., desktop PC, laptop, smartphone) and usage log (e.g., active vs. idle).

3.3. Preliminary study for scenario identification

We selected an office scenario for data collection due to its relevance as one of the common environments and the lack of research on user modeling in office settings, despite its importance for context-aware assistant systems (Yan and Selker, 2000). This preliminary study aimed to specify suitable office scenarios for data collection.

Participants and Procedures: Eleven participants (7 males, 4 females; $M = 27.09$, $SD = 1.58$) from a local university community in South Korea voluntarily took part. On average, they spent 8 hours/day in the office ($M = 8.00$, $SD = 1.28$). Participants were asked to provide details about their daily office routines and tasks, responding to an open-ended prompt: “Please describe a series of actions you do while in the office, specifically with whom and what you are doing.” This provided insights into typical office behaviors.

Qualitative Results: We performed qualitative analysis, analyzing 340 words of subjective responses using QDA Miner Lite,¹ following methodology from Lu and Bowman (2021), Saldaña (2021). Responses were coded and categorized, yielding 77 valid data points ($M = 7.00$, $SD = 2.57$). The distribution was slightly right-skewed ($skewness = 0.84$), but showed no substantial distortion or outliers.

Analysis identified six main categories: **individual work** (e.g., check schedule) (48.05%), **individual break** (e.g., watch videos) (9.09%),

individual walk (e.g., move to other rooms) (9.09%), **collaborative work** (e.g., attend meetings) (20.78%), **collaborative break** (e.g., chat with colleagues) (7.79%), and **collaborative walk** (e.g., move with colleagues) (5.19%). Individual work was most frequent, while collaborative walk was least. These findings highlight key behavioral patterns: working vs. resting, solitary vs. social activities, and stationary vs. mobile tasks.

3.4. Data collection and user feedback

3.4.1. System implementation for dataset acquisition

We developed a system for dataset acquisition to construct our user model. Table 3 details the collected sensor and interaction data. Hardware used for dataset collection included the HoloLens 2,² the Empatica E4 wristband,³ a Windows 11 desktop PC, an ASUS Zenbook laptop, and a Samsung Galaxy S20+ smartphone.

Physical condition. We collected inertial measurement unit (IMU) sensor data from HoloLens 2 (3-axis acceleration (ACC): 10 Hz, 3-axis angular velocity (GYRO): 20 Hz, 3-axis magnetic field (MAG): 5 Hz) and Empatica E4 (3-axis acceleration (ACC): 32 Hz).

Cognitive-emotional status. Empatica E4 collected electrodermal activity (EDA) (4 Hz), blood volume pulse (BVP) (64 Hz), heart rate (HR) (1 Hz), and skin temperature (TEMP) (4 Hz). These measurements are automatically preprocessed and computed outputs by the E4’s proprietary algorithm. In particular, the photoplethysmography (PPG) signals are processed to derive interbeat interval (IBI) (Hu et al., 2024), from which HR is computed and provided by the device.

E4 wristband was used for physiological data collection, consistent with prior research leveraging its capabilities for cognitive load and emotion recognition. In particular, several studies have provided datasets using E4 for emotional valence detection (Park et al., 2020; Saganowski et al., 2022), often with reported success (Zitouni et al., 2021; Subathra and Malarvizhi, 2023). Yet, emotion recognition is highly subject-dependent (Mattern et al., 2023; Bulagang et al., 2021), requiring multiple validation methods and careful generalization.

Social circumstances. Face detection data was collected via HoloLens 2 camera sensor (30 Hz) using the UWP FaceAnalysis API.⁴ Ambient audio was recorded at 48 kHz.

¹ QDA Miner Lite: <https://provalisresearch.com/products/qualitative-data-analysis-software/>.

² Microsoft HoloLens 2: <https://www.microsoft.com/en-us/hololens>.

³ Empatica E4: <https://www.empatica.com/en-int/research/e4/>.

⁴ <https://github.com/microsoft/Windows-universal-samples/tree/main/Samples/HolographicFaceTracking>

Table 3
Overview of collected dataset.

Modeling dimension	User Factor	Device	Data source	Raw data	Unit	Sampling rate
Physical	Posture, Energy expenditure	HoloLens 2	IMU	3-axis Acceleration (a_x, a_y, a_z)	(m/s^2)	10 Hz
				3-axis Angular Velocity ($\omega_x, \omega_y, \omega_z$)	(rad/s)	20 Hz
				3-axis Magnetic Field (m_x, m_y, m_z)	(μT)	5 Hz
Cognitive-Emotional	Cognitive Load, Emotional Arousal, Emotional Valence	Empatica E4	IMU	3-axis Acceleration ($\alpha_x, \alpha_y, \alpha_z$)	(g)	32 Hz
		Empatica E4	EDA	EDA	(μS)	4 Hz
			TEMP	TEMP	($^{\circ}\text{C}$)	4 Hz
			PPG	BVP	–	64 Hz
Social	Social Engagement	HoloLens 2	Camera	Number of Faces (n_{faces})	–	30 Hz
			Microphone	Audio Amplitude (MFCCs)	–	48 kHz
Behavioral	Device type, Usage Log	Window 11 Desktop PC	Mouse Keyboard Activity Log	Activity Indicator {0, 1}	–	3 Hz
		ASUS Zenbook Laptop	Mouse Keyboard Activity Log	Activity Indicator {0, 1}	–	3 Hz
		Samsung Galaxy S20+ Smartphone	Screen On/Off	Activity Indicator {0, 1}	–	3 Hz

Device interaction behavior. Activity logs were recorded from a Windows 11 desktop, ASUS Zenbook laptop, and Samsung Galaxy S20+. Mouse/keyboard activity was logged at 3 Hz, while smartphone screen status (on/off) was recorded at 3 Hz (0 = idle, 1 = active).

Data communication. IMU data from the HoloLens 2 was transmitted via User Datagram Protocol (UDP) socket communication through the local WiFi network. HoloLens 2 was remotely controlled through the Windows Device Portal. Data from E4 was monitored through the E4 real-time streaming app⁵ using Samsung Galaxy S20+ and accessed via the Empatica Connect web platform, E4 manager, over Bluetooth⁶.

3.4.2. Office setup and procedures

Hardware: The office setup included a Windows 11 desktop PC, an ASUS Zenbook laptop, a Samsung Galaxy S20+ smartphone, and a desk with a chair. Participants wore an E4 on their left wrist and a HoloLens 2 throughout the experiment. Participants were allowed to freely use a desktop PC, a laptop, and a smartphone as needed.

Procedures: Data was collected using a semi-naturalistic approach inspired by Bao and Intille (2004), allowing greater behavioral variability than strictly controlled environments. Participants performed typical office activities in a lab-based office setup, following an opportunistic sensing paradigm (Lane et al., 2008) to minimize intervention and better capture authentic user behavior.

Participants completed six tasks, each followed by a 10 min break, designed to simulate real-world office scenarios (Fig. 3).

- Individual Work (IW): Reading documents, checking the calendar, and programming.

- Individual Break (IB): Watching videos, listening to music, and using a smartphone.
- Individual Walk (IWk): Walking alone within the office.
- Collaborative Work (CW): Work-related discussions and brainstorming with the experimenter, who acted as a colleague.
- Collaborative Break (CB): Casual, non-work-related conversations (e.g., about the weather or hobbies) with the experimenter.
- Collaborative Walk (CWk): Walking with the experimenter to simulate a casual walk with a colleague.

To capture ground truth on cognitive-emotional states, participants used a virtual interface on HoloLens 2 to report their cognitive load (high/medium/low), emotional arousal (high/medium/low), and emotional valence (positive/neutral/negative) for each task, using in-situ natural labeling based on the Experience Sampling Method (ESM) (van Berkel et al., 2017).

After data collection, participants provided open-ended feedback on integrating user models into context-aware AR interfaces. We aimed to identify user expectations and preferences for an adaptive AR glass interface, informing future adaptive AR development with our user model.

Participants: Twelve participants (7 males, 5 females; $M = 28.08$, $SD = 1.56$) from a local university community in South Korea took part. On a 5-point Likert scale, they reported an average AR/VR technology frequency score of 3.5 ($SD = 1.31$) and experience score of 4.08 ($SD = 0.79$), with all having used AR glasses at least once. The study was voluntary and approved by the Institutional Review Board (IRB).

3.5. Data preparation

This section outlines data consolidation, pre-processing, separation, and feature extraction. The overall process is illustrated in Fig. 4.

⁵ <https://play.google.com/store/apps/details?id=com.empatica.e4realtime>

⁶ <https://e4.empatica.com/connect>

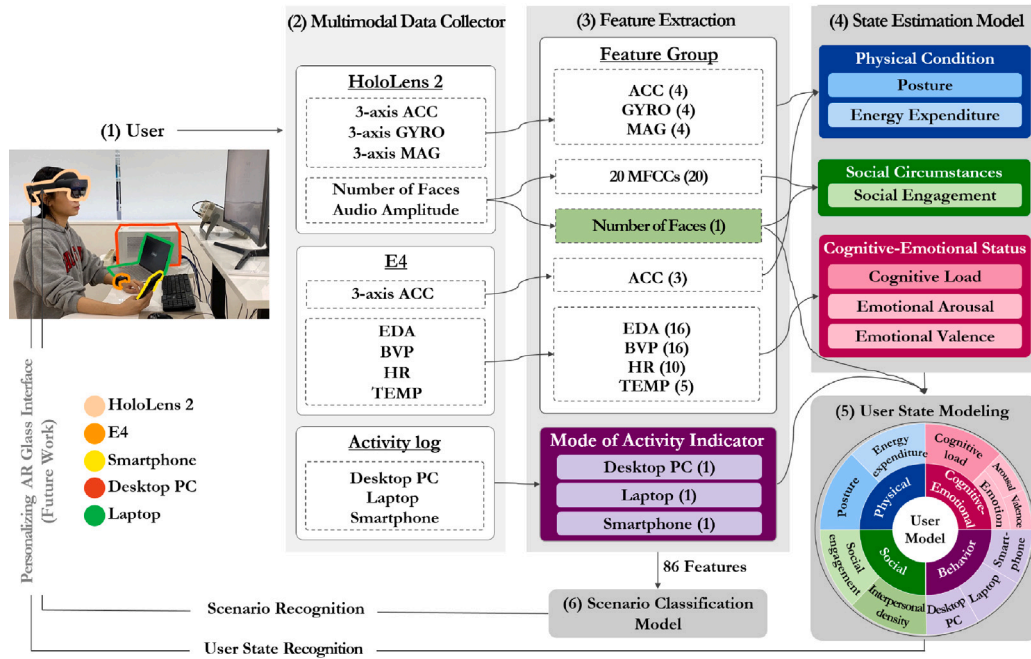


Fig. 4. Overall procedure of data process for user state inference and scenario classification. The number in the parentheses denotes the number of features.

3.5.1. Data consolidation

We collected approximately 648 min of data (9 min $\times$ 6 conditions $\times$ 12 participants). Each device's data was stored as timestamped numpy arrays and merged using an outer join, retaining all time points while filling missing values with NA. Audio data was processed using librosa (v.0.10.0.post2) (McFee et al., 2015) to extract 20 Mel-frequency cepstral coefficients (MFCCs). Missing values were handled via linear interpolation or nearest neighbor methods, depending on the data properties.

3.5.2. Pre-processing

We applied pre-processing and signal filtering to remove noise, artifacts, and outliers.

For EDA signals, noise, artifact and outlier removal methods were adapted from Taylor et al. (2015). A 1 Hz low-pass filter was applied to remove high-frequency noise. Artifact removal involved threshold-based filtering ($\geq 20\%$ increase, $\leq 10\%$ decrease) (Kocielnik et al., 2013), exponential decay verification for unnatural signal decay, and motion artifact detection using concurrent EDA and ACC changes. Outlier removal addressed quantization error (amplitude discontinuities exceeding 5%) and physiologically invalid skin conductance level (SCL) values (≤ 0). NA values were interpolated using linear interpolation.

For BVP signals, despite E4's embedded motion artifact removal (Cosoli et al., 2021; Poli et al., 2023), we conducted supplementary pre-processing to reduce any additional noise and artifact, adapting the approach from Vollmer (2019). A median filter was used to remove spike noise and smooth the signal (Chandra et al., 2021), followed by a fourth-order Butterworth bandpass filter (1 Hz–8 Hz) to eliminate irrelevant frequencies (Heinisch et al., 2018). Wrist motion artifacts were removed by detecting simultaneous abrupt changes in ACC and R-R intervals derived from the BVP signal. NA values were interpolated using linear interpolation.

For HR signals, the E4's built-in algorithm automatically computes HR from interbeat interval (IBI) data. E4 is known as a reliable tool as the proprietary algorithm removes unwanted motion artifact (Chandra et al., 2021), preprocesses PPG signals to derive IBI (Hu et al., 2024), and removes wrong beats before exporting IBI (Milstein and Gordon, 2020). To ensure data integrity, HR values were screened for deviations beyond 40–200 bpm (Chung et al., 2019; Mejía-Mejía and Kyriacou,

2023), and the 20% rule (Collins et al., 2019) was applied to detect outliers based on consecutive HR change rates. No outliers were detected in the dataset.

TEMP signals were screened for values outside 29–39 °C and abrupt fluctuations exceeding 20% between consecutive samples. NA values were interpolated using median. A Kalman filter was applied to ACC, GYRO, and MAG signals from HoloLens 2 and from the E4 to reduce noise.

3.5.3. Data separation

To prevent data leakage, data separation was strictly performed before applying the sliding window and feature extraction. The datasets were partitioned into train and test sets depending on each validation method, preserving the temporal order within each participant and condition.

For hold-out, we segmented each participant-condition pair, maintaining the original temporal order: the first 80% for train, and the remaining 20% for test. These segments were then aggregated to form the train and test sets. This approach ensures balanced distribution of participants and conditions and temporal continuity of the physiological signals without random sampling or shuffling.

For within-participant, data were split similarly as in hold-out, per participant and condition, with the first 80% for train and the last 20% for test, but the resulting segments were grouped separately for each participant to form individual training and test sets. The segmentation procedures for hold-out and within-participant are illustrated in Fig. 5.

For 5-fold cross, we applied StratifiedGroupKFold from scikit-learn (v.1.2.2) to partition the dataset, ensuring (1) balanced distribution of conditions across folds, and (2) that data from the same participant were kept within a single fold to prevent overlap between training and test sets. Lastly, for Leave-One-User-Out (LOUO), each participant was assigned to the test set in turn.

3.5.4. Feature extraction

After the dataset was split into train and test set, the data was segmented using a sliding window technique (Banos et al., 2014) in its train or test set. Each window spanned 320 data points with a 50% overlap, a strategy widely used in prior studies (Trethewey, 2000; Bao and Intille, 2004).

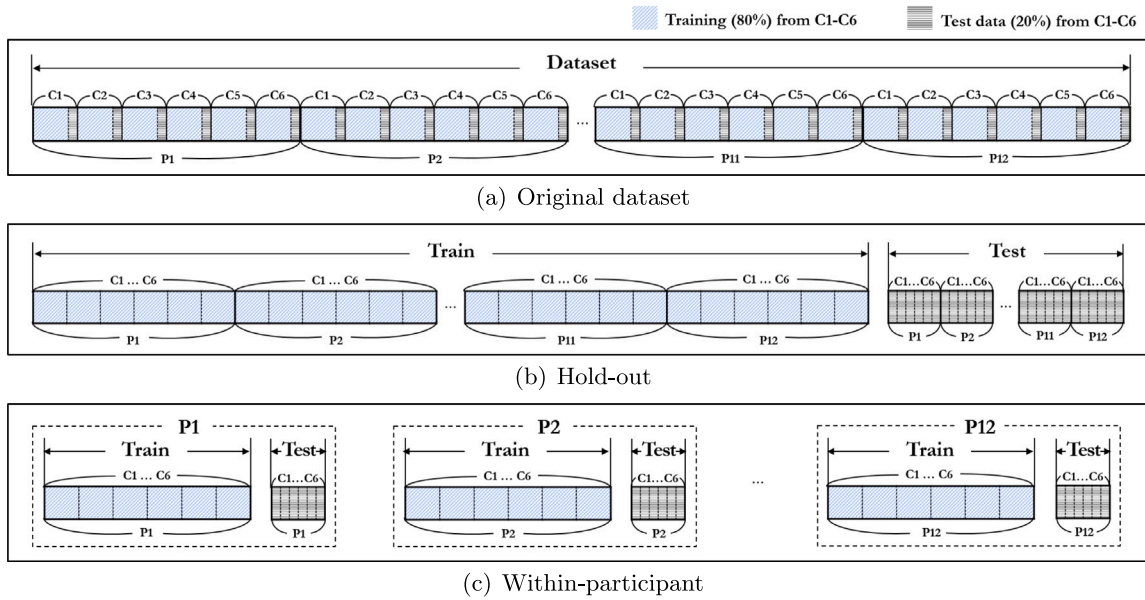


Fig. 5. Illustration of separation procedures: (a) original dataset, (b) hold-out, and (c) within-participant.

Feature extraction was performed on physiological signals (EDA, BVP, HR, TEMP), motion signals (ACCs, GYRO, MAG), and contextual data (n_{faces} , MFCCs, device activity indicators). For frequency domain feature extraction, power spectral density was computed using Welch's method. In total, 86 features were extracted (Table A.2).

Among 86 features, 47 physiological features were considered as candidates for cognitive-emotional status estimation. To identify the most influential features, feature importance was computed separately for cognitive load, arousal, and valence using the Mean Decrease in Impurity (MDI) based on Gini impurity. The top 20 features for each three targets were selected and the three subsets were merged with duplicates removed to form the final input set. For example, the combined feature set included 22 features under the hold-out.

All continuous features were standardized using a StandardScaler (scikit-learn (v.1.2.2)) (Pedregosa et al., 2011). To prevent any potential data leakage by ensuring that the test set does not influence the scaling process, the scaler was fitted on the training set and applied to the test set.

3.6. Estimation model development

We developed a neural network to estimate user states requiring inference, including posture, energy expenditure, cognitive load, emotional arousal, emotional valence, and social engagement. Interpersonal density and device interaction behavior were directly mapped to user states. Additionally, a scenario classification model was built to distinguish between six scenarios.

We employed Random Forest (100 trees, max depth=3) and Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997), the latter being effective in handling time-series data over time. LSTM networks were chosen for their effectiveness in user locomotion prediction (Mayor et al., 2024) and emotion classification (Zitouni et al., 2021).

The model training was implemented using the TensorFlow (v2.10.1) (Abadi et al., 2016) and scikit-learn (v1.2.2) (Pedregosa et al., 2011). Fig. 6 illustrates the model architecture. Appendix Table A.3 details state estimation models, while Appendix Table A.4 presents the scenario classification model.

Cognitive load, emotional arousal, and emotional valence were annotated across real-world settings, resulting in class distributions that reflect typical imbalances commonly reported in prior studies (Schmidt

Table 4
Distribution of levels for cognitive load, emotional arousal and valence.

Level	Cognitive load	Emotional arousal	Emotional valence
Low	15	4	8
Medium/Neutral	29	36	52
High	28	32	12
Total	72	72	72

et al., 2018; Nandi et al., 2021). The distribution of levels is presented in Table 4.

Since time-series data contain temporally structured information, simple augmentation techniques such as oversampling, random transformations, or synthetic generation are inappropriate, as they risk distorting underlying temporal patterns (Iwana and Uchida, 2021). Therefore, we adopted focal loss (Lin et al., 2017) for LSTM models and class weighting for Random Forest models. These strategies help balanced learning by reducing bias toward majority classes.

To address class imbalance, we applied class-aware weight strategies in both LSTM and Random Forest models. For the LSTM models, we used focal loss with alpha values derived from the observed class distribution. For cognitive load, which showed moderate class imbalance, we applied a mild adjustment by assigning weights based on one minus each class's relative frequency. For emotional arousal and valence, where class proportions were more uneven, weights were assigned in inverse proportion to class frequencies to better represent minority classes.

In the Random Forest models, global class weights for cognitive load were computed once based on the inverse of the class frequencies, whereas for arousal and valence, tree-specific weights were computed per tree based on the class distribution within its bootstrap sample.

3.7. Validation methods

The model was assessed using accuracy, precision, recall, and F1 score. Hold-out assessed overall performance, 5-fold cross measured generalization, LOUO evaluated user dependency, and within-participant validation examined intra-participant consistency and time dependency.

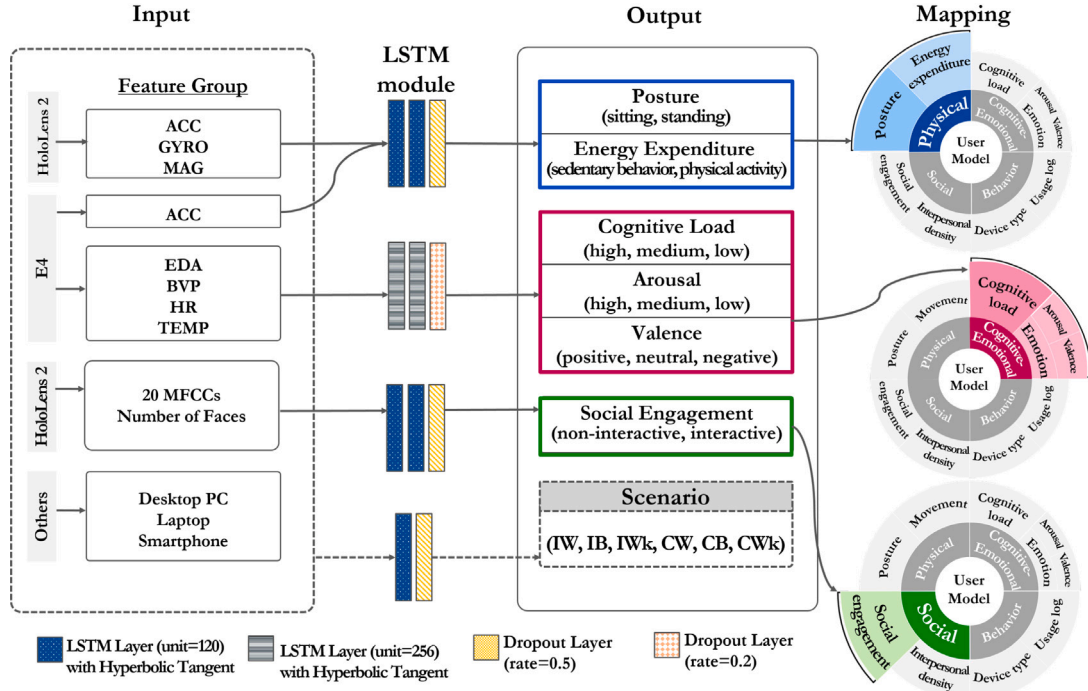


Fig. 6. A pipeline for estimating user states and classifying scenarios.

Table 5

Performance metrics for state estimation models using LSTM and Random Forest under hold-out validation.

Model	Method	Accuracy(%)	Precision(%)	Recall(%)	F1 Score(%)
Posture	LSTM	97.75	97.75	97.75	97.75
	Random Forest	95.54	95.67	95.54	95.47
Energy Expenditure	LSTM	97.78	97.78	97.78	97.78
	Random Forest	95.54	95.67	95.54	95.47
Social Engagement	LSTM	99.40	99.40	99.40	99.40
	Random Forest	99.40	99.41	99.40	99.40
Cognitive Load	LSTM	77.48	78.57	77.48	77.79
	Random Forest	72.42	73.85	72.42	72.92
Emotional Arousal	LSTM	79.17	79.53	79.17	79.08
	Random Forest	70.21	72.02	70.21	70.58
Emotional Valence	LSTM	78.57	80.42	78.57	79.24
	Random Forest	73.74	78.62	73.74	75.40

4. Results

We primarily report accuracy, with precision, recall, and F1 score included in tables.

4.1. States estimation

Table 5 compares LSTM and Random Forest models using hold-out validation, while Table 6 reports LSTM results for 5-fold, LOUO, and within-participant validation.

Physical Condition Estimation: The posture and energy expenditure estimation models performed consistently across validation strategies. Posture estimation achieved 97.75% (hold-out), 94.73% (5-fold), 93.98% (LOUO), and 98.74% (within) accuracy. Energy expenditure estimation reached 97.78% (hold-out), 94.71% (5-fold), 93.93% (LOUO), and 98.74% (within) accuracy.

Cognitive-Emotional Status Estimation: Cognitive load, emotional arousal, and emotional valence models performed best in

within-participant validation (above 90% accuracy). Cognitive load estimation achieved 77.48% (hold-out), 70.54% (5-fold), 63.82% (LOUO), improving to 91.96% (within) accuracy. Emotional arousal estimation reached 79.17% (hold-out), 72.95% (5-fold), 65.56% (LOUO), and 92.56% (within) accuracy. Emotional valence estimation attained 78.57% (hold-out), 69.00% (5-fold), 65.48% (LOUO), with 93.65% (within) accuracy.

We compared our LSTM models with a majority class baseline that always predicts the most frequent label. With the class imbalance mitigation strategy, our models improve balanced accuracy from the majority-class baseline of 33% to 73%–75%, and similarly increase macro F1 from $\leq 28\%$ to $\geq 70\%$ (Table 7). Balanced accuracy and macro-F1 reflect performance on minority classes by averaging metrics across classes, reducing the influence of majority class dominance.

Table 8 further shows that recall and F1 score for the minority classes consistently exceed 60%, confirming that the model does not ignore underrepresented labels. These results demonstrate that the network effectively learns all classes rather than defaulting to the majority class.

Table 6

Performance metrics for state estimation models using LSTM under 5-fold, LOUO, and within-participant validations.

Model	Validation	Accuracy(%)	Precision(%)	Recall(%)	F1 score(%)
Posture	5-fold	94.73	94.82	94.73	94.72
	LOUO	93.98	94.37	93.98	94.00
	Within	98.74	98.77	98.74	98.74
Energy Expenditure	5-fold	94.71	94.81	94.71	94.70
	LOUO	93.93	94.33	93.93	93.96
	Within	98.74	98.77	98.74	98.74
Social Engagement	5-fold	97.36	97.39	97.36	97.36
	LOUO	95.85	96.14	95.85	95.84
	Within	99.37	99.39	99.37	99.37
Cognitive Load	5-fold	70.54	75.37	70.54	71.25
	LOUO	63.82	75.04	63.82	66.23
	Within	91.96	93.03	91.96	92.01
Emotional Arousal	5-fold	72.95	74.83	72.95	72.62
	LOUO	65.56	75.64	65.56	65.68
	Within	92.56	93.84	92.56	92.66
Emotional Valence	5-fold	69.00	77.24	69.00	71.75
	LOUO	65.48	70.45	65.48	66.64
	Within	93.65	94.74	93.65	93.71

Table 7

Balanced accuracy and macro F1 comparison between the baseline (majority class predictor) and our model.

Model	Balanced accuracy (%)		Macro F1 (%)	
	Baseline	Ours	Baseline	Ours
Cognitive Load	33.33	75.37	19.16	75.15
Emotional Arousal	33.33	74.07	22.22	75.79
Emotional Valence	33.33	73.01	27.96	70.08

Social Circumstance Estimation: The social engagement model performed strongly, achieving 99.40% (hold-out), 97.36% (5-fold), 95.85% (LOUO), and 99.37% (within) accuracy.

Comparison with Random Forest Baseline: Compared to LSTM models, the Random Forest baseline performed similarly in social engagement (99.40%), but slightly lower in posture and energy expenditure (95.54%). It showed notably lower accuracy in cognitive load (72.42%), emotional arousal (70.21%), and emotional valence (73.74%).

Post-hoc Analysis: SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017) was used to analyze feature contributions. SHAP values were normalized to sum to 100% for better interpretability, with results in Table 9.

4.2. Scenario classification

Results for LSTM and Random Forest under hold-out validation are shown in Table 10, while LSTM results under 5-fold, LOUO, and within-participant validations are provided in Table 11. The LSTM model outperformed Random Forest in hold-out validation (99.01% vs. 92.36% accuracy) and achieved 79.19% (5-fold), 81.68% (LOUO), and 99.50% (within) accuracy.

To assess device dependency, we excluded desktop PC, laptop, and smartphone activity indicator from scenario classification. Notably, hold-out accuracy remained at 97.92%, suggesting the model is not heavily reliant on device activity data and can adapt to alternative device sources.

4.3. Feedback

The first question was: (1) “Which one do you prefer between a context-aware AR glass in which the interface is adjusted depending on your state or a conventional static AR glass interface adjusted manually?” 11 out of 12 participants preferred an adaptive AR interface, favoring automatic adjustments over manual control.

The second question was: (2) “Do you think that a user model is necessary to guide the placement of a virtual interface in AR?” Again, 11 participants agreed that a user model is essential for effective virtual content placement in AR.

The third question was: (3) “Which specific user factors can be utilized to guide the placement?”; The fourth was: (4) “We are in the process of developing a user model with user-relevant factors for AR glass users. What are your expectations for how it could enhance AR interactions?”

For the third and fourth questions, we conducted thematic analysis inspired by Braun and Clarke (2006) to categorize key insights. We reviewed all feedback, grouped similar responses, assigned codes to highlight key aspects, and identified common themes, excluding responses irrelevant to our research focus.

As a result, we identified four key response themes (R1-R4):

R1. Adapting AR Visuals Based on Cognitive-Emotional Cue: Participants emphasized adjusting opacity and detail levels based on mental load. P4,9: “It would be nicer if the opacity and level of detail are controlled depending on the user’s mental load”. They also favored content that could improve mood when stressed. P6,10: “I would appreciate the virtual content recommendation that could steer me toward a positive mood when detecting I am stressed out”.

R2. Social Interaction Enhancement Through Contextual AR: Enhancing social interaction without causing disruptions was deemed crucial. P6: “It would be helpful if the system helps my social interaction skills. But it is as much as important not to intervene the social interaction by placing it in a non-intrusive manner”. They also suggested collaborative tools for meetings. P9: “When involving meetings or group communications, suggesting virtual contents like sticky notes that multiple users can interact with simultaneously would be beneficial”.

R3. Timing of AR Interface Transitions: Participants preferred adjustments at major task or temporal transitions rather than reacting to every minor state changes. P5,8: “It would be nicer for the system

Table 8

Level-wise recall and F1 scores for cognitive load, emotional arousal, and emotional valence.

Metric	Level	Cognitive load	Emotional arousal	Emotional valence
Recall (%)	Low	64.29	62.50	66.96
	Medium/Neutral	83.50	84.92	82.42
	High	78.32	74.78	69.64
F1 Score (%)	Low	60.95	67.96	62.24
	Medium/Neutral	80.62	80.30	85.78
	High	83.88	79.10	62.23

Table 9

Top three normalized SHAP values for state estimation models.

Posture		Energy Expenditure		Social Engagement	
Feature	Value (%)	Feature	Value (%)	Feature	Value (%)
GYRO intensity	19.99	GYRO intensity	15.56	MFCC1 mean	19.14
MAG intensity	15.16	MAG intensity	14.08	n _{faces} mode	12.01
ACC.Z mean (E4)	13.24	ACC.X mean	12.59	MFCC7 mean	10.23
Cognitive Load		Emotional Arousal		Emotional Valence	
Feature	Value (%)	Feature	Value (%)	Feature	Value (%)
BVP std	8.27	BVP median	10.84	BVP median	17.29
HR wavelet energy	7.89	BVP total power	8.75	BVP max	10.08
HR median	7.02	BVP max	8.68	BVP std	7.71

Table 10

Performance metrics for scenario classification using LSTM and Random Forest under hold-out validation.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
LSTM	99.01	99.01	99.01	99.01
Random Forest	92.36	92.81	92.36	92.36

Table 11

Performance metrics for scenario classification using LSTM under 5-fold, LOUO, and within-participant validations.

Validation	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
5-fold	79.19	81.58	79.19	77.68
LOUO	81.68	84.38	81.68	79.54
Within	99.50	99.52	99.50	99.50

to execute adjustments at major task transitions or substantial temporal points”.

R4. Mixed-Initiative Systems: Participants highlighted the importance of mixed-initiative systems where control is shared between the system and the user (Findlater and McGrenere, 2004). P1,3,5,6,7: “An adaptive interface is beneficial, but it would be enhanced if the user could also use explicit voice commands or keywords, similar to Siri”. P4: “Upon detecting changes in user state, it is better for the system to first seek user permission before making interface adjustments”. P11: “A consistent and designated area for interaction, such as the wrist or peripheral vision, could be used to query user consent; following approval, the interface could be transferred”.

5. Discussion

5.1. Analysis

RQ1 (Key User Factors for Wearable AR): To identify key factors, we defined modeling dimensions (physical, cognitive-emotional, social, behavioral) and explored relevant literature. This led to designing the HQS-AR model, also supported by participant feedback. R1 findings emphasized adjusting opacity, detail, and contents based on cognitive

load and emotions, while R2 findings highlighted the importance of non-intrusive AR interfaces that enhance social engagement.

RQ2 (Estimating User Factors from Multimodal Sensor Data):

We collected motion, visual, audio, behavioral, and physiological data to develop machine learning models inferring posture, energy expenditure, cognitive load, emotional arousal, emotional valence, social engagement, and scenario. Performance was high for posture, energy expenditure, social engagement, and scenario classification, while cognitive-emotional models showed lower accuracy in LOUO but improved significantly in within-participant validation, highlighting the need for personalization due to individual physiological variations.

RQ3 (Considerations for User-Adaptive AR Systems):

Opened feedback confirmed that user expectations align with the identified factors from the literature survey. Participants also preferred interface transitions during major task shifts or prolonged states rather than minor fluctuations, emphasizing the balance between automation and system stability.

Additionally, users favored a mixed-initiative approach over full automation, preferring confirmation prompts or explicit initiation. This aligns with prior findings (Lu and Xu, 2022), where semi-automated AR transitions were most effective and preferred. These insights clarify the optimal timing for adaptations and underscore the importance of integrating the mixed-initiative approach into the future adaptive AR systems.

5.2. Comparison to related work using physiological data

5.2.1. Cognitive load

To contextualize our work, Table 12 provides an overview of key findings from related studies using multimodal physiological data for cognitive load estimation. While our model does not achieve the highest accuracy, direct comparisons are limited due to several key differences.

Task design and data collection: Unlike prior studies conducted in controlled VR settings (Collins et al., 2019; Souchet et al., 2022), our study was conducted in a semi-naturalistic, lab-based AR office environment that allowed naturalistic interactions, making the experimental setting fundamentally different. While controlled experiments optimize accuracy, real-world variability enhances ecological validity for practical AR applications.

Table 12

Overview of related studies using physiological data for cognitive load (CL) or mental workload classification, based on selected aspects.

	Collins et al. (2019)	Luong et al. (2020)	Souchet et al. (2022)	Bhat et al. (2023)
Aim	Develop ERVE methodology to measure CL objectively in VR learning	Recognize mental workload in VR using customized HMD with physiological sensors	Classify CL in VR and assess psychophysiological impact under Stroop task	Classify CL in VR training using EEG and EDA
Dataset	Experimentally collected from 4D hypercube rotation puzzle task in VR (N=24)	Experimentally collected from a VR flight simulator task (N=75)	Experimentally collected under Stroop task (N=92)	Experimentally collected during LEGO task in VR (N=25)
Modality	Heart Activity	Ocular Activity, EDA, Cardiac Activity	Eye-tracking, Electrocardiogram (ECG), EDA	EEG, EDA
Class	3	4	3	2
Validation	10-fold	10-fold	70/30 train-test split	Nested k-fold cross (k=10)
Performance	91.75% (accuracy)	65% (accuracy)	87.23% (accuracy)	80.4% (accuracy)

Dataset characteristics (modality, class, and dataset size): Sensor modalities, class structures, and dataset size differ across studies. For example, Souchet et al. (2022), Bhat et al. (2023) used ECG or EEG, which we did not use, and Bhat et al. (2023) performed binary classification, whereas we adopted a multi-class approach. Dataset size also varied, limiting direct comparisons. If we expand our dataset, we expect further improvements in accuracy.

5.2.2. Emotion

To contextualize our work, Table 13 presents an overview of related studies on emotion classification using multimodal physiological data. As with cognitive load, differences in datasets and classification types make direct comparisons difficult. Below, we outline key insights from related work.

Binary classification and subject dependency: High-performing studies (Zitouni et al., 2021; Mattern et al., 2023) conducted binary arousal-valence classification but suggested subject-dependent variation. Mattern et al. (2023) also showed lower performance in LOSO validation, suggesting that physiological variability might drive subject dependency.

Four-quadrant classification and individual variability: Expanding to four-quadrant classification, Bulang et al. (2021) achieved high accuracy in intra-subject validation but lower accuracy in inter-subject validation, suggesting that individual differences in physiological responses may limit cross-subject generalizability.

5.3. Application

Using State Indicators for Virtual Configuration: Our user model is expected to serve as a reference for context-aware AR systems. Physical indicators can guide spatial layout to avoid obstructing the user's movement, while cognitive load levels can inform adjustments of content density to avoid overload. Emotional states can guide content recommendations, while social indicators can be used to trigger subtle interaction and prevent face occlusion. Device activity logs allow for more personalized suggestions and assistive displays.

Modeling User Routine Through Patterns: By aggregating state indicators over time, patterns in indicators may suggest user's daily routines. Consistent patterns in physical, cognitive-emotional, social, and behavioral cues often reflect recurring tasks. For example, a stationary posture, high cognitive load, low social engagement, and active desktop

PC usage each morning may indicate individual work. Fig. 7 illustrates how patterns of state indicators can be mapped to six scenarios.

To test this, we trained a model using the state indicators to classify six scenarios. With the actual labeled state, accuracy reached about 87%, but using the inferred states from raw sensor data reduced accuracy to about 63%, likely due to compounded errors in multi-step inference. This highlights the need for precise state estimation before applying the state indicators for routine task prediction.

Application Scenarios: We illustrate how our user model can enhance wearable AR interfaces through personalization using an office worker persona. A visual example is provided in Appendix Fig. A.1.

During individual work, the system configures productivity-enhancing virtual widgets in a world-fixed coordinate system. If mental tension is detected (high cognitive load or stress), it suggests simplifying the interface and provides stress-relief content. When the user leaves for lunch, virtual content is minimized to prevent distractions while walking.

During casual chatting, the system activates face occlusion avoidance and subtle gesture interactions, and suggests relevant content to support the conversation. In a work meeting, it facilitates collaboration by recommending productivity tools and reducing unimportant notifications to avoid disruption.

These scenarios illustrate how our user model can be used to dynamically adapt AR interfaces, reducing the need for manual adjustments while enhancing usability. We envision our user model being applied to wearable AR applications, contributing to the seamless integration of AR technology into daily life.

5.4. Limitations and future work

Future research could expand our model by considering static attributes (e.g., age, height, gender, personality) alongside the dynamic profiling (Farseev et al., 2016; Eke et al., 2019) we focused on. These factors could enrich user understanding and further refine AR personalization. Additionally, our model remains adaptable, allowing for sub-component modifications based on emerging evidence of important user factors and varying application domains.

Expanding to diverse environments, refining task distinctions, and collecting longitudinal data could enhance the applicability and temporal insight across diverse contexts in real-world AR scenarios. While we focused on an office setting, extending to home or industrial environments may improve generalizability. Refining scenario categories could

Table 13
Overview of related studies using physiological data for emotion classification, based on selected aspects. LOSO stands for Leave-One-Subject-Out validation.

	Zitouni et al. (2021)	Mattern et al. (2023)	Bulagang et al. (2021)	Siirtola et al. (2023)
Aim	Classify emotions in arousal-valence space using peripheral physiological signals	Explore emotion recognition beyond labs using wrist-worn physiological signals	Investigate if HR signals alone can classify four-class emotions	Predict emotions using wrist-worn biosignals and evaluate different model performance
Dataset	K-EmoCon dataset (N=20 chosen) (Park et al., 2020)	Experimentally collected under autobiographical emotion memory task (N=40)	Experimentally collected under 16 short VR videos (N=20)	WESAD (N=15) (Schmidt et al., 2018)
Modality	HR, EDA, TEMP	BVP, EDA, TEMP, ACC	HR	EDA, BVP, HR, HRV, TEMP
Class	2 classes each for arousal and valence	2 classes each for arousal and valence	4 (four quadrants based on valence and arousal)	Continuous
Validation	4-fold cross	10-fold cross, LOSO	Intra-subject, Inter-subject (10-fold cross)	LOSO
Performance	Arousal: 90.79%/Valence: 90.53% (accuracy)	[10-fold] Arousal: 85%, Valence: 80%/[LOSO] Arousal: 52%, Valence: 55% (accuracy)	[Intra] highest: 100%, lowest: 36.3%, [Inter]: 46.7% (accuracy)	Arousal: 0.59 (MSE), 0.81 (R^2)/Valence: 0.43 (MSE), 0.71 (R^2)

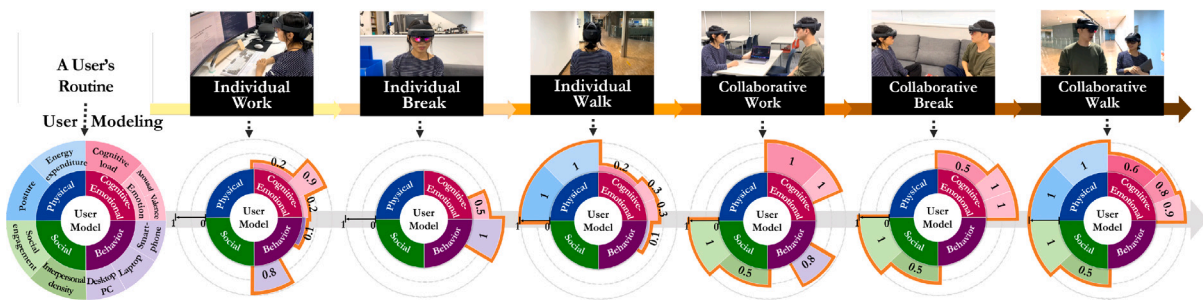


Fig. 7. Exemplary mapping of state patterns across six scenarios, providing a unified representation for each scenario. The states are quantified from 0 to 1, where 0 indicates an inactive or low-intensity state, and 1 indicates an active or higher-intensity state.

enable more nuanced AR adaptations, and longitudinal data would support a deeper understanding of user states over time.

A key future direction is the real-time implementation of our model, including both real-time state inference and adaptive content delivery. While we developed machine learning algorithms for state and scenario inference, real-time deployment was beyond the scope of this study. Future research could validate how real-time state inference and interface adaptation based on our model enhance user experience. We believe our findings contribute to establishing a foundation for these advancements.

6. Conclusion

This study introduced Holistic Quantified-Self for wearable AR (HQS-AR), a unified user model for AR glass users, incorporating key factors across physical, cognitive-emotional, social, and behavioral dimensions. We developed machine learning algorithms for state

estimation (approximately 88% accuracy) and scenario classification (approximately 99% accuracy). Participant feedback confirmed the alignment of the proposed user model with user expectations and highlighted considerations for designing adaptive AR interfaces. By proposing a reference user state model, we hope to contribute to the development of personalized AR systems that align more closely with users' needs.

CRedit authorship contribution statement

Eunhwa Song: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Taewook Ha:** Software, Methodology, Investigation, Data curation, Conceptualization. **Junhyeok Park:** Software, Methodology, Investigation, Formal analysis, Data curation. **Hyunjin Lee:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Woontack Woo:** Writing – review

& editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Fig. A.1 and Tables A.1–A.4.

Table A.1

Overview of prior studies, aligned with the problem spaces and parameters defined in this study.

Modeling dimension	Component	Problem space	Parameter	Related studies
Physical Condition	Posture, Energy Expenditure	What	Contents	Tsai and Huang (2018) Shakeri et al. (2023)
			Level of detail	Tsai and Huang (2018)
		Where	Position	Tsai and Huang (2018) Klose et al. (2019) Lages and Bowman (2019a) Lages and Bowman (2019b) Cheng et al. (2023) Lee and Woo (2023) Lee et al. (2023a)
			Coordinate system	Klose et al. (2019) Lages and Bowman (2019b) Fukushima et al. (2020) Lu and Bowman (2021) Lee and Woo (2023) Lee et al. (2023a)
		How	Occlusion avoidance	Tsai and Huang (2018)
		When	Timing	Makolkina and Paramonov (2020)
Cognitive-Emotional Status	Cognitive load	What	Level of detail	Julier et al. (2000) DiVerdi et al. (2004) Tatzgern et al. (2016) Lindlbauer et al. (2019)
	Emotion	What	Contents	Chatzopoulos and Hui (2016) Ronquillo et al. (2020)
Social Circumstances	Social engagement, interpersonal density	What	Contents	Ajanki et al. (2010) Davari et al. (2022)
		Where	Position	Orlosky et al. (2015) Janaka et al. (2022) Medeiros et al. (2022)
		How	Occlusion avoidance	Orlosky et al. (2015) Davari et al. (2022) Medeiros et al. (2022)
			Mode of interaction	Lee et al. (2017) Lee et al. (2018b) Lee et al. (2019)
Device Interaction Behavior	Usage log, device type	What	Contents	Ho et al. (2011) Liang et al. (2019) Lee et al. (2023b) Shakeri et al. (2023)
		Where	Position	Feiner et al. (1997) Millette and McGuffin (2016) Langner et al. (2021) Bang and Woo (2023)

Table A.2
List of extracted features and description.

Feature group	Domain	Feature type	Description
EDA	Time (Posada-Quintero et al., 2016; Shukla et al., 2021; Kong et al., 2023)	Mean, Min, Max, Std, Median	
		Skin Conductance Level (SCL)	Mean skin conductance level
		Non-Specific SCR (NS-SCR)	Number of non-specific skin conductance responses
		SCR Amplitude	Mean amplitude of SCR peaks
		SCR Frequency	Frequency of SCR occurrences (per minute)
		SCR Latency	Time to first SCR peak (seconds)
		SCR Rise Time	Mean rise time from SCR onset to peak
	Frequency (Shukla et al., 2021; Huang et al., 2024)	High-Frequency Power	High frequency (0.15–0.4 Hz) power of EDA signal
		Total Power	Total power of EDA signal in frequency domain
		Dominant Frequency	Dominant frequency with highest power in EDA signal
BVP	Wavelet (Shukla et al., 2021; Huang et al., 2024)	Wavelet Energy	Wavelet energy of approximation coefficients
		Wavelet Entropy	Wavelet entropy of approximation coefficients
	Time (Chowdhury et al., 2019; Giannakakis et al., 2022; Stuyck et al., 2022; Campanella et al., 2023)	Mean, Min, Max, Std, Median	
		RR Intervals (Mean)	Mean RR interval between BVP peaks
		RR Intervals (Std)	Standard deviation of RR intervals
		SDNN	Standard deviation of RR intervals
		RMSSD	Root mean square of successive RR interval differences
		NN50	Count of RR interval differences > 50 ms
		PNN50	Percentage of NN50 over total RR intervals
	Frequency (Schmidt et al., 2018; Giannakakis et al., 2022; McNaboe et al., 2022; Campanella et al., 2023)	Total Power	Total power of BVP signal in frequency domain
		High-Frequency Power	High frequency (0.15–0.4 Hz) power of BVP signal
		Dominant Frequency	Dominant frequency with highest power in BVP signal
	Wavelet (Bračić and Stefanovska, 1998)	Wavelet Energy	Wavelet energy of approximation coefficients in BVP signal
		Wavelet Entropy	Wavelet entropy of approximation coefficients in BVP signal
HR	Time (Chowdhury et al., 2019)	Mean, Min, Max, Std, Median	
	Frequency (Schmidt et al., 2018; Chowdhury et al., 2019; Collins et al., 2019; Giannakakis et al., 2022; McNaboe et al., 2022)	Total Power	Total power of HR signal in frequency domain
		High-Frequency Power	High frequency (0.15–0.4 Hz) power of HR signal
		Dominant Frequency	Dominant frequency with highest power in HR signal
	Wavelet (Rajendra Acharya et al., 2015)	Wavelet Energy	Wavelet energy of approximation coefficients in HR signal
		Wavelet Entropy	Wavelet entropy of approximation coefficients in HR signal
TEMP	Time (Chowdhury et al., 2019)	Mean, Min, Max, Std, Median	
ACC from E4	Time	Mean per ACC axis (X, Y, Z)	
ACC, GYRO, MAG from HoloLens 2	Time (Bao and Intille, 2004; Yang et al., 2008; Steven Eyobu and Han, 2018)	Mean per axis (X, Y, Z) for each ACC, GYRO, MAG Magnitude of each ACC, GYRO, MAG	$acc_{mag} = \sqrt{a_x^2 + a_y^2 + a_z^2}$ $gyr_{mag} = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$ $mag_{mag} = \sqrt{m_x^2 + m_y^2 + m_z^2}$
MFCCs	Time (McFee et al., 2015)	Mean	Mean of MFCC coefficient 0–19
Face	Time	Mode	Mode of face detection status
Desktop, laptop, smartphone	Time	Mode	Mode of usage status

Table A.3
Overview of model architectures for state estimations.

Components	Physical condition	Cognitive-emotional status	Social circumstance
Input Features	Mean and magnitude of ACC, GYRO, MAG from HoloLens 2; Mean ACC from E4.	Combined set of selected features from EDA, BVP, HR, TEMP feature group	Mean of 20 MFCCs, mode of n_{faces} .
LSTM Layers	Two LSTM layers (120 units) with tanh activation.	Two LSTM layers (256 units) with tanh activation.	Two LSTM layers (120 units) with tanh activation.
Dropout	Dropout layer with a rate of 0.5 post-LSTM to prevent overfitting.	Dropout layer with a rate of 0.2 post-LSTM to prevent overfitting.	Dropout layer with a rate of 0.5 post-LSTM to prevent overfitting.
Output Layers	Two sigmoid layers to predict posture and energy expenditure.	Three softmax layers for multi-class classification: cognitive load, emotional arousal, emotional valence.	Sigmoid layer for binary classification.

Table A.4
Overview of model architecture for scenario classification.

Component	Description
Input Features	Mean and magnitude of ACC, GYRO, MAG from HoloLens 2, mean of ACC from E4, mean of 20 MFCCs, mode of n_{faces} , 47 features from EDA, BVP, HR, TEMP feature group, mode of activity indicator.
LSTM Layer	One LSTM layer (120 units) with tanh activation.
Dropout	Dropout rate of 0.5 post-LSTM to prevent overfitting.
Output Layer	Softmax layer with 6 units for multi-class classification.



(a): Mental Tension



(b): Work Meeting

Fig. A.1. Illustration of two exemplary application scenarios. Once the system identifies the user's state and scenario, the AR interface adapts accordingly to assist the user.

Data availability

Data will be made available on request.

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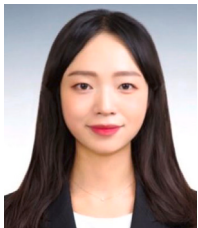
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